

Task 6: Create a Strong Password and Evaluate Its Strength

Objective

To understand the characteristics of a strong password, evaluate multiple passwords using an online password strength checker, and learn best practices to protect against common password attacks.

Methodology

- 1. Created multiple sample passwords with varying complexity.
- 2. Tested each password on passwordmeter.com.
- 3. Recorded strength scores and tool feedback.
- 4. Summarized observations and extracted best practices.
- 5. Researched common password attacks and analyzed the role of password complexity.

Password Testing & Results

Password	Score (out of 100)	Feedback Summary
password123	43	Too common, dictionary word, predictable sequence.
Pass@123	77	Better with mixed case and symbol, but too short.
Mys3cureP@ss	88	Stronger, mixes upper/lowercase, numbers, symbols.
T!9uR\$e4LxQw	100	Very strong: long, random, complex mix of characters.

Screen Shots

task 6[1].pdf

Password strength evaluation

passwordmeter.com - Search

Password Strength Checker

https://passwordmeter.com

The Password Meter

Test Your Password

Hide: ☒

Score: 100%

Complexity: Very Strong

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	12	+ 48
Uppercase Letters	Cond/Incr	$++((len-n)*2)$	4	+ 16
Lowercase Letters	Cond/Incr	$++((len-n)*2)$	4	+ 16
Numbers	Cond	$+(n*4)$	2	+ 8
Symbols	Flat	$+(n*6)$	2	+ 12
Middle Numbers or Symbols	Flat	$+(n*2)$	4	+ 8
Requirements	Flat	$+(n*2)$	5	+ 10

Deductions

Letters Only	Flat	-n	0	0
Numbers Only	Flat	-n	0	0
Repeat Characters (Case Insensitive)	Comp	-	0	0

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Password Strength Meter

passwordmonster.com

PasswordMonster

info@passwordmonster.com

How Secure is Your Password?

Take the Password Test

Tip: Stronger passwords use different types of characters

Show password: ☒

T!9uR\$e4LxQw

Very Strong

12 characters containing: Lower case Upper case Numbers Symbols

Time to crack your password:
4 million years

Review: Fantastic, using that password makes you as secure as Fort Knox.

indeed

Want to stand out? Prioritize your visibility on the homepage of relevant job seekers

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https://passwordmeter.com

The Password Meter

Test Your Password

Password: password123

Hide: ☒

Score: 43%

Complexity: Good

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	11	+ 44
Uppercase Letters	Cond/Incr	$+\frac{((len-n)*2)}{4}$	0	0
Lowercase Letters	Cond/Incr	$+\frac{((len-n)*2)}{4}$	8	+ 6
Numbers	Cond	$+(n*4)$	3	+ 12
Symbols	Flat	$+(n*6)$	0	0
Middle Numbers or Symbols	Flat	$+(n*2)$	2	+ 4
Requirements	Flat	$+(n*2)$	3	0
Deductions				
Letters Only	Flat	-n	0	0
Numbers Only	Flat	-n	0	0
Repeat Characters (Case Insensitive)	Comp	-	2	- 2

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https://passwordmeter.com

The Password Meter

Test Your Password

Password: Mys3cureP@ss

Hide: ☒

Score: 88%

Complexity: Very Strong

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	12	+ 48
Uppercase Letters	Cond/Incr	$+\frac{((len-n)*2)}{4}$	2	+ 20
Lowercase Letters	Cond/Incr	$+\frac{((len-n)*2)}{4}$	8	+ 8
Numbers	Cond	$+(n*4)$	1	+ 4
Symbols	Flat	$+(n*6)$	1	+ 6
Middle Numbers or Symbols	Flat	$+(n*2)$	2	+ 4
Requirements	Flat	$+(n*2)$	5	+ 10
Deductions				
Letters Only	Flat	-n	0	0
Numbers Only	Flat	-n	0	0
Repeat Characters (Case Insensitive)	Comp	-	3	- 2

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The screenshot shows the Password Strength Checker interface. The password 'Pass@123' is entered, and the score is 77%, categorized as 'Strong'. The interface includes a table of additions and deductions for password strength.

Test Your Password		Minimum Requirements	
Password:	Pass@123	- Minimum 8 characters in length - Contains 3/4 of the following items: - Uppercase Letters - Lowercase Letters - Numbers - Symbols	
Hide:	☒		
Score:	77%		
Complexity:	Strong		

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	8	+ 32
Uppercase Letters	Cond/Incr	$+\frac{(len-n)*2}{1}$	1	+ 14
Lowercase Letters	Cond/Incr	$+\frac{(len-n)*2}{3}$	3	+ 10
Numbers	Cond	$+(n*4)$	3	+ 12
Symbols	Flat	$+(n*6)$	1	+ 6
Middle Numbers or Symbols	Flat	$+(n*2)$	3	+ 6
Requirements	Flat	$+(n*2)$	5	+ 10

Deductions	Type	Rate	Count	Bonus
Letters Only	Flat	$-n$	0	0
Numbers Only	Flat	$-n$	0	0
Repeat Characters (Case Insensitive)	Comp	-	2	- 2
Consecutive Uppercase Letters	Flat	$-(n*2)$	0	0
Consecutive Lowercase Letters	Flat	$-(n*2)$	2	- 4
Consecutive Numbers	Flat	$-(n*2)$	2	- 4

Key Findings

- **Length matters:** Short passwords (<8 characters) are weak even if they include symbols.
- **Complexity helps:** Mixing uppercase, lowercase, numbers, and symbols increases strength.
- **Avoid dictionary words:** Common words (e.g., “password”, “welcome”) are easily guessed by attackers.
- **Randomness > Patterns:** Sequential or predictable patterns (e.g., 1234, abcd) weaken security.
- **Best combination:** Long (12+ characters), random, and mixed-character passwords are most secure.

Common Password Attacks

- **Brute Force:** Automated tools try all possible combinations; longer, complex passwords take exponentially more time to crack.
- **Dictionary Attacks:** Attackers use precompiled lists of common words/passwords; avoiding dictionary words prevents easy compromise.
- **Credential Stuffing:** Attackers reuse leaked credentials across platforms; using unique passwords per site is critical.

Best Practices for Strong Passwords

1. Use at least **12–16 characters**.
2. Mix **uppercase, lowercase, numbers, and symbols**.
3. Avoid dictionary words, names, or predictable patterns.
4. Prefer **randomly generated passphrases** (e.g., Cr!ms0n_Star!89) over single words.
5. Use a **password manager** to store unique strong passwords for each account.
6. Enable **multi-factor authentication (MFA)** wherever possible.

Conclusion

From the evaluation, the strongest tested password (T!9uR\$e4LxQw) scored 100/100, showing that randomness, length, and character diversity are the keys to strength. Understanding password attacks highlights the importance of complexity and uniqueness. By following best practices, users can significantly reduce the risk of their accounts being compromised.